

DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA

MESTRADO EM ENGENHARIA DE COMPUTADORES E TELEMÁTICA

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MODELAÇÃO E DESEMPENHO DE REDES E SERVIÇOS

PRACTICAL GUIDE NO. 2

Task 1

Consider the discrete event simulator, implemented in the provided MATLAB function *Sim2Flows*, to estimate the performance of a point-to-point link supporting 2 packet flows. The simulator considers a link of capacity C (in Mbps) and a queue of size f (in Bytes) with a FIFO (First-In-First-Out) scheduling discipline.

The packet Flow 1 is characterized by: (i) a packet arrival following a Poisson process with average λ_1 pps and (ii) a random packet size between 64 and 1518 bytes with the probabilities: 19% for 64 bytes, 23% for 110 bytes, 17% for 1518 bytes and an equal probability for all other values (i.e., from 65 to 109 and from 111 to 1517).

The packet Flow 2 is characterized by: (i) a packet arrival following a Poisson process with average λ_2 pps and (ii) a random packet size between 110 and 130 bytes (all values with equal probability).

Input parameters of *Sim2Flows*:

- λ – vector with the packet rate of each flow, in pps
- C – link capacity, in Mbps
- f – queue size, in Bytes
- P – total number of transmitted packets of a simulation run

Performance parameters estimated by *Sim2Flows*:

- PL – vector with the Packet Loss (in %) of each flow
- APD – vector with the Average Packet Delay (in milliseconds) of each flow
- MPD – vector with the Maximum Packet Delay (in milliseconds) of each flow
- TT – vector of the Transmitted Throughput (Mbps) of each flow

Stopping criterion of *Sim2Flows*:

- Time instant when the link finishes the transmission of the Pth packet (P is the 4th input parameter); in *Sim2Flows*, the queued packets at the end of the simulation do not count for performance estimation.

1.a. Develop a MATLAB script to run *Sim2Flows* 50 times with a stopping criterion of P = 10.000 at each run and to compute the estimated values and the 90% confidence intervals of all performance parameters when λ_1 = 1500 pps, λ_2 = 1000 pps, C = 10 Mbps and f = 1.000.000 Bytes (~1 MByte). Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.72e+00 +- 7.47e-02
  Max. Packet Delay (ms) = 1.50e+01 +- 7.38e-01
  Throughput (Mbps)      = 7.45e+00 +- 3.29e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.32e+00 +- 7.30e-02
  Max. Packet Delay (ms) = 1.45e+01 +- 7.08e-01
  Throughput (Mbps)      = 9.63e-01 +- 3.84e-03
```

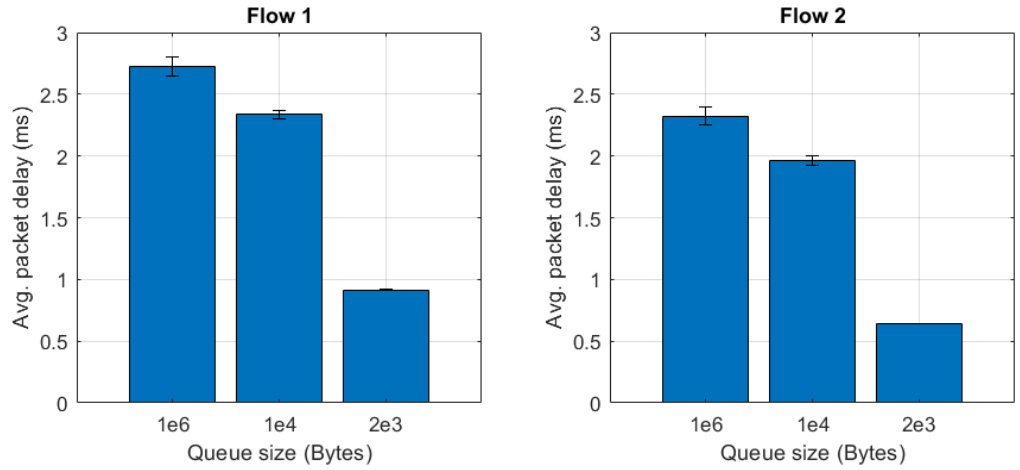
1.b. Repeat experiment **1.a** now considering f = 10.000 Bytes (~10 KBytes). Justify the differences between these results and the results of experiment **1.a**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)         = 4.45e-01 +- 4.02e-02
  Avg. Packet Delay (ms)  = 2.33e+00 +- 3.67e-02
  Max. Packet Delay (ms)  = 8.81e+00 +- 3.41e-02
  Throughput (Mbps)       = 7.38e+00 +- 2.97e-02
Flow 2:
  Packet Loss (%)         = 5.99e-02 +- 1.26e-02
  Avg. Packet Delay (ms)  = 1.96e+00 +- 3.67e-02
  Max. Packet Delay (ms)  = 8.45e+00 +- 5.77e-02
  Throughput (Mbps)       = 9.61e-01 +- 3.23e-03
```

1.c. Repeat experiment **1.a** now considering f = 2.000 Bytes (~2 KBytes). Justify the differences between these results and the results of experiments **1.a** and **1.b**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)         = 9.49e+00 +- 8.40e-02
  Avg. Packet Delay (ms)  = 9.16e-01 +- 3.68e-03
  Max. Packet Delay (ms)  = 2.69e+00 +- 1.23e-02
  Throughput (Mbps)       = 6.09e+00 +- 1.69e-02
Flow 2:
  Packet Loss (%)         = 1.29e+00 +- 5.29e-02
  Avg. Packet Delay (ms)  = 6.41e-01 +- 3.10e-03
  Max. Packet Delay (ms)  = 2.56e+00 +- 1.46e-02
  Throughput (Mbps)       = 9.43e-01 +- 3.01e-03
```

1.d. Present the simulation results concerning the average packet delay obtained in the previous experiments of in a bar chart with the confidence intervals in error bars on the same plot [Footnote 1]. Analyse the width of the obtained confidence intervals and take conclusions on the confidence of the results. The visualization should look like:



Footnote 1 https://www.mathworks.com/help/matlab/creating_plots/bar-chart-with-error-bars.html

Task 2

In the previous Task 1, the provided MATLAB function *Sim2Flows* was used to estimate the performance of a point-to-point link supporting 2 packet flows statistically multiplexed in a single queue. In this Task 2, the aim is to estimate the performance parameters considering a link with priority queueing where one of the flows has the highest priority and the other has the lowest priority.

First, develop a MATLAB function with name *Sim2FlowsPrio* with the appropriate discrete event simulator (use the provided MATLAB function *Sim2Flows* as starting point of the new simulator's development). In *Sim2FlowsPrio*:

- the input parameter *f* should now be a vector with the size (in Bytes) of each queue;
- a new input parameter must be added, whose value is either 1 (if Flow 1 has higher priority than Flow 2) or 2 (if Flow 2 has either priority than Flow 1);
- all other input parameters remain the same;
- all output parameters remain the same.

2.a. Develop a MATLAB script to run *Sim2FlowsPrio* 50 times with a stopping criterion of P = 10.000 at each run and to compute the estimated values and the 90% confidence intervals of all performance parameters when Flow 2 has higher priority than Flow 1, λ_1 = 1500 pps, λ_2 = 1000 pps, C = 10 Mbps, $f_1 = f_2$ = 1.000.000 Bytes. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.91e+00 +- 7.16e-02
  Max. Packet Delay (ms) = 1.59e+01 +- 5.97e-01
  Throughput (Mbps)      = 7.46e+00 +- 3.01e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 4.88e-01 +- 2.42e-03
  Max. Packet Delay (ms) = 1.38e+00 +- 6.41e-03
  Throughput (Mbps)      = 9.60e-01 +- 3.52e-03
```

2.b. Repeat experiment **2.a** now considering that Flow 1 has higher priority than Flow 2. Justify the differences between these results and the results of experiment **1.a** and **2.a**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 1.86e+00 +- 3.56e-02
  Max. Packet Delay (ms) = 1.11e+01 +- 5.16e-01
  Throughput (Mbps)      = 7.44e+00 +- 2.75e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 8.58e+00 +- 4.66e-01
  Max. Packet Delay (ms) = 7.93e+01 +- 4.60e+00
  Throughput (Mbps)      = 9.60e-01 +- 4.15e-03
```

2.c. Repeat experiment **2.a** now considering $f_1 = f_2 = 10.000$ Bytes. Justify the differences between these results and the results of experiment **2.a**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 4.38e-01 +- 4.08e-02
  Avg. Packet Delay (ms) = 2.54e+00 +- 4.35e-02
  Max. Packet Delay (ms) = 9.90e+00 +- 5.73e-02
  Throughput (Mbps)      = 7.39e+00 +- 3.08e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 4.84e-01 +- 2.34e-03
  Max. Packet Delay (ms) = 1.38e+00 +- 7.41e-03
  Throughput (Mbps)      = 9.59e-01 +- 3.10e-03
```

2.d. Repeat experiment **2.b** now considering $f_1 = f_2 = 10.000$ Bytes. Justify the differences between these results and the results of experiment **2.b**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 1.39e-01 +- 1.89e-02
  Avg. Packet Delay (ms) = 1.79e+00 +- 2.14e-02
  Max. Packet Delay (ms) = 8.57e+00 +- 6.60e-02
  Throughput (Mbps)      = 7.40e+00 +- 2.54e-02
Flow 2:
  Packet Loss (%)        = 1.15e-01 +- 5.94e-02
  Avg. Packet Delay (ms) = 7.82e+00 +- 2.92e-01
  Max. Packet Delay (ms) = 6.94e+01 +- 3.64e+00
  Throughput (Mbps)      = 9.56e-01 +- 3.61e-03
```

Task 3

In the previous Task 2, you have developed the MATLAB function *Sim2FlowsPrio* implementing a discrete event simulator to estimate the performance parameters considering a link with priority queueing. In this Task 3, the aim is to estimate the performance parameters considering a link whose packet scheduling algorithm is the DRR (Deficit Round Robin).

First, develop a MATLAB function with name *Sim2FlowsDRR* with the appropriate discrete event simulator (use the MATLAB function *Sim2FlowsPrio* developed in Task 2 as starting point of the new simulator's development). In *Sim2FlowsDRR*:

- a new vector input parameter must be added, with the Quantum value (in Bytes) assigned to each flow;
- a new Boolean parameter *Inv* must be added, whose value is FALSE (if each round-robin round is by the order of the flow ID) or TRUE (if each round-robin round is by the inverse order of the flow ID);
- the previous input parameter defining the highest priority flow must be eliminated;
- all other input parameters remain the same;
- all output parameters remain the same.

3.a. Develop a MATLAB script to run *Sim2FlowsDRR* 50 times with a stopping criterion of P = 10.000 at each run and to compute the estimated values and the 90% confidence intervals of all performance parameters when the Quantum values are 1600 for Flow 1 and 1600 for Flow 2, *Inv* = FALSE, λ_1 = 1000 pps, λ_2 = 5500 pps, C = 10 Mbps, $f_1 = f_2$ = 1.000.000 Bytes. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.42e+01 +- 4.02e+00
  Max. Packet Delay (ms) = 5.97e+01 +- 6.49e+00
  Throughput (Mbps)      = 4.84e+00 +- 2.43e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.56e+01 +- 2.62e+00
  Max. Packet Delay (ms) = 5.80e+01 +- 4.59e+00
  Throughput (Mbps)      = 5.10e+00 +- 1.68e-02
```

3.b. Repeat experiment **3.a** now considering *Inv* = TRUE. Take conclusions on the differences between these results and the results of experiment **3.a**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.62e+01 +- 3.78e+00
  Max. Packet Delay (ms) = 6.30e+01 +- 6.17e+00
  Throughput (Mbps)      = 4.84e+00 +- 2.52e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.67e+01 +- 3.11e+00
  Max. Packet Delay (ms) = 6.02e+01 +- 5.59e+00
  Throughput (Mbps)      = 5.10e+00 +- 1.73e-02
```

3.c. Repeat experiment **3.a** now considering the Quantum values of 1600 for Flow 1 and 4800 for Flow 2. Justify the differences between these results and the results of experiment **3.a**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 4.65e+01 +- 5.93e+00
  Max. Packet Delay (ms) = 1.11e+02 +- 1.08e+01
  Throughput (Mbps)      = 4.66e+00 +- 1.34e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 3.36e+00 +- 3.15e-02
  Max. Packet Delay (ms) = 6.25e+00 +- 1.43e-01
  Throughput (Mbps)      = 5.28e+00 +- 1.33e-02
```

3.d. Repeat experiment **3.d** now considering *Inv* = TRUE. Take conclusions on the differences between these results and the results of experiment **3.c**. Results (recall that these are simulation results):

```
Flow 1:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 5.16e+01 +- 4.65e+00
  Max. Packet Delay (ms) = 1.12e+02 +- 7.90e+00
  Throughput (Mbps)      = 4.67e+00 +- 1.57e-02
Flow 2:
  Packet Loss (%)        = 0.00e+00 +- 0.00e+00
  Avg. Packet Delay (ms) = 2.18e+00 +- 1.84e-02
  Max. Packet Delay (ms) = 4.81e+00 +- 1.14e-01
  Throughput (Mbps)      = 5.27e+00 +- 1.42e-02
```